

$$(2x_0)$$

15

C

At resonance, $x_0 = 16.4$ cm and $f = 1.5$ Hz (read from graph)

Maximum acceleration $a_0 = \omega^2 x_0 = (2\pi f)^2 x_0 = (2\pi \times 1.5)^2 (0.164) = 14.6 \text{ m s}^{-2}$

The fact that the graph does not go to infinity at resonance shows that there is damping.